

University Interscholastic League
2023 – 2024 Elementary Number Sense Test A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.
Stop – Wait for Signal!

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| <p>(1) $6 + 41 =$ _____</p> <p>(2) $18 + 29 =$ _____</p> <p>(3) $76 - 32 =$ _____</p> <p>(4) $224 \div 2 =$ _____</p> <p>(5) $13 \times 4 =$ _____</p> <p>(6) $204 - 31 =$ _____</p> <p>(7) $9 + 11 + 13 =$ _____</p> <p>(8) $43 \times 11 =$ _____</p> <p>(9) $832 \div 8 =$ _____</p> <p>*(10) $3897 + 703 =$ _____</p> <p>(11) $25 \times 32 =$ _____</p> <p>(12) Which digit is in the hundreds place in 49317.20568? _____</p> <p>(13) $15 \times 22 =$ _____</p> <p>(14) 83670.2874 rounded to the tens place is _____</p> <p>(15) What is the remainder for $60318 \div 5$? _____</p> <p>(16) The number of whole numbers between 103 and 125 is _____</p> <p>(17) $6 \times 10^{-1} + 5 \times 10^{-2} + 1 \times 10^{-3} =$ _____ (decimal)</p> <p>(18) $33 \times 7 - 33 \times 5 =$ _____</p> <p>(19) MDCLX = _____ (Arabic Numeral)</p> | <p>*(20) $320 \times 375 =$ _____</p> <p>(21) $8 + 4 \div 2 =$ _____</p> <p>(22) $16 + 14 + 12 + 10 + 8 =$ _____</p> <p>(23) 72 hours = _____ days</p> <p>(24) Which is larger: 0.1 or $\frac{1}{9}$? _____</p> <p>(25) $\frac{7}{12} + \frac{1}{12} =$ _____</p> <p>(26) $19 \times 21 =$ _____</p> <p>(27) 44% = _____ common fraction</p> <p>(28) If 24 ♥ costs 44¢, then 36 ♥ cost _____ ¢</p> <p>(29) $21 \div \frac{7}{12} =$ _____</p> <p>*(30) $443 \times 359 =$ _____</p> <p>(31) 16 quarters = _____ dimes</p> <p>(32) The sum of the two smallest primes greater than 30 is _____</p> <p>(33) 2 quarts = _____ ounces</p> <p>(34) $1025 \div 25 =$ _____</p> <p>(35) $16\frac{2}{3}\%$ = _____ common fraction</p> <p>(36) The GCF of 24 and 32 is _____</p> <p>(37) $97 \times 95 =$ _____</p> |
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- (38) $\frac{1}{12} + \frac{11}{12} - \frac{3}{12} =$ _____
- (39) The ratio, in cents, of 2 quarters to \$2 is _____
- *(40) $225 \times 439 =$ _____
- (41) $14^2 =$ _____
- (42) $6^3 =$ _____
- (43) The volume of a cube with edge 4 is _____
- (44) The perimeter of an isosceles triangle with congruent sides 18 and other side 24 is _____
- (45) $2^4 + 3^2 =$ _____
- (46) $2\frac{1}{8} \times 6\frac{1}{8} =$ _____ (mixed number)
- (47) What is the volume of a rectangular box that is 8-ft by 6-ft by 11-ft? _____ cubic feet
- (48) $37.5\% \times 64 =$ _____
- (49) If $x = \frac{2}{3}$, then $21 + 3x =$ _____
- *(50) $2023 + 2024 + 2025 =$ _____
- (51) What is the number, **k**, in the sequence: 0, 1, 8, **k**, 64, 125, ...? _____
- (52) $9\frac{4}{5} + 6\frac{9}{10} =$ _____ (mixed number)
- (53) If the radius of a circle is 48, then the circumference of the circle is _____ π
- (54) If a rectangle with area 24 has a width that measures 4, then its perimeter is _____
- (55) 23 (base 10) = _____ (base 3)
- (56) The number of days in September is _____
- (57) What is the area of a triangle with altitude $4\frac{1}{2}$ and base 24? _____
- (58) How many elements are in the power set for $\{0, 2, 4, 6, 8\}$? _____
- (59) If three times a whole number, minus nine, is thirty-three, what is the number? _____
- *(60) 150 yards = _____ inches
- (61) 1011 (base 2) = _____ (base 8)
- (62) $12 + -2^4 \times (-4) =$ _____
- (63) What is the probability a blue marble is randomly picked from a bag of 6 blue and 8 red marbles? _____
- (64) 25 cups = _____ ounces
- (65) $72^2 =$ _____
- (66) $9^6 \div 7$ has remainder of _____
- (67) The area of a rhombus with diagonals 18 and 24 is _____
- (68) If $18 - 3x > 54$, then $x <$ _____
- (69) $2\frac{1}{42} - \frac{7}{6} =$ _____
- *(70) $\sqrt{130321} =$ _____
- (71) The additive inverse of $6\frac{1}{2}$ is _____
- (72) $88^2 - 12^2 =$ _____
- (73) If $9\frac{3}{4}\%$ of x is $3\frac{1}{4}\%$ of 6, then $x =$ _____
- (74) $1^0 + 2^1 + 3^2 + 4^3 =$ _____
- (75) $33^2 + 11^2 =$ _____
- (76) $(2 + 4 + 6 + \dots + 10)^2 =$ _____
- (77) What is the area of a right triangle with hypotenuse 13 and leg 5? _____
- (78) $\sqrt{576} \times \sqrt{625} =$ _____
- (79) $24 + 24^2 =$ _____
- *(80) $25^4 =$ _____

2023 – 2024 University Interscholastic League Elementary Number Sense Test A – Key

(1) 47	*(20) 114000 – 126000	(38) $\frac{3}{4}$; .75	(59) 14
(2) 47	(21) 10		*(60) 5130 – 5670
(3) 44	(22) 60	(39) $\frac{1}{4}$; .25	(61) 13
(4) 112	(23) 3		(62) 76
(5) 52	(24) $\frac{1}{9}$	*(40) 93837– 103713	(63) $\frac{3}{7}$
(6) 173		(41) 196	(64) 200
(7) 33	(25) $\frac{2}{3}$	(42) 216	(65) 5184
(8) 473	(26) 399	(43) 64	(66) 1
(9) 104		(44) 60	(67) 216
*(10) 4370 – 4830	(27) $\frac{11}{25}$	(45) 25	(68) -12
(11) 800	(28) 66	(46) $13\frac{1}{64}$	(69) $\frac{6}{7}$
(12) 3	(29) 36	(47) 528	*(70) 343 – 379
(13) 330	*(30) 151086 – 166988	(48) 24	(71) $-6\frac{1}{2}$; -6.5; $-\frac{13}{2}$
(14) 83670	(31) 40	(49) 23	(72) 7600
(15) 3	(32) 68	*(50) 5769 – 6375	(73) 2
(16) 21	(33) 64	(51) 27	(74) 76
(17) .651	(34) 41	(52) $16\frac{7}{10}$	(75) 1210
(18) 66	(35) $\frac{1}{6}$	(53) 96	(76) 900
(19) 1660	(36) 8	(54) 20	(77) 30
	(37) 9215	(55) 212	(78) 600
		(56) 30	(79) 600
		(57) 54	*(80) 371094 – 410156
		(58) 32	

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.